Visual Analytics Homework
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1 Classification of Visual Tasks, Data Types, Relationships, and Visual Variables

In this section, we classify the visualizations presented above, considering both the individual (Federico) and group (4 students) cases.

1.1 Individual Performance: Federico

1.1.1 Bar Chart of Federico's Grades

- Visual Task: Comparison – allows comparison of grades across exams.
- Type of Data:
  - Nominal: exam names.
  - Quantitative: numeric grades.
  - Ordinal/Derived: honors (30L) highlighted.
  - Temporal: exams are sorted chronologically.
- Relationships:
  - Each exam corresponds to one grade.
  - Honors (30L) are visually distinguished from normal grades.
- Visual Variables / Variations:

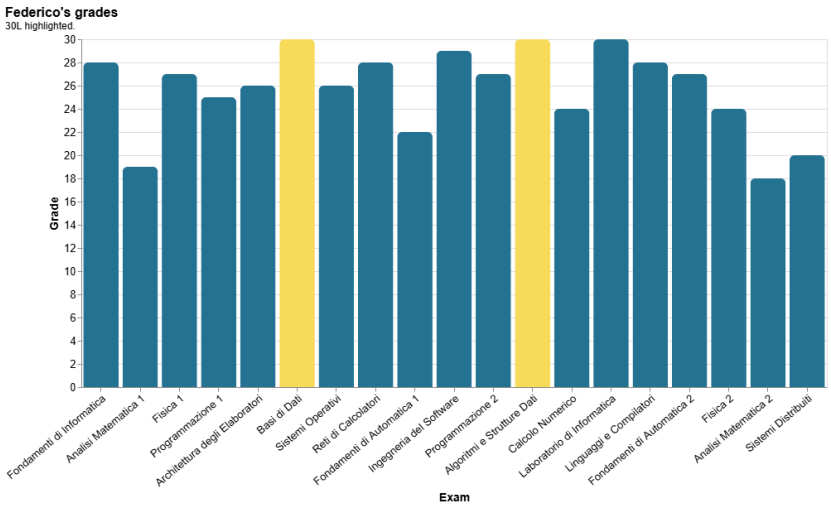


Figure 1: Bar Chart for Federico's Grades.

– Position (x-axis): ordinal, exams.
– Position (y-axis): quantitative, grade value.
– Color: nominal, highlights 30L.

1.1.2 Temporal Trend of Federico's Grades

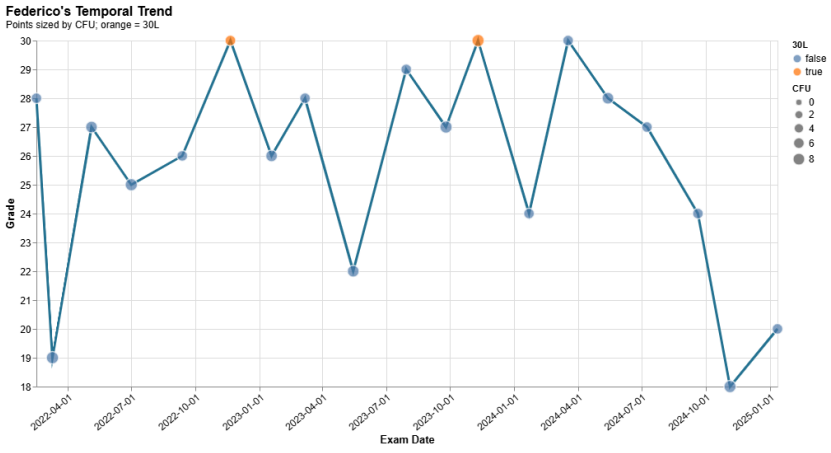


Figure 2: Line Chart for Federico's Grades.

- Visual Task: Trend / Change over Time – shows how grades evolve temporally.
- Type of Data:
  - Temporal: exam dates.
  - Quantitative: numeric grades, CFU (point size).
  - Nominal: honors (30L).
- Relationships:
  - Exam date → grade (temporal progression).
  - CFU → point size.
  - Honors → point color (e.g., orange for 30L).
- Visual Variables / Variations:
  - Position (x-axis): temporal (exam date).
  - Position (y-axis): quantitative (grade).
  - Color: nominal (30L vs normal).
  - Size: quantitative (CFU).
  - Line / Point Overlay: shows individual temporal trends.

1.2 Group Performance: 4 Students

1.2.1 Box Plot of Group Grades

- Visual Task: Comparison and Distribution – compares grades across students and shows spread.
- Type of Data:
  - Nominal: student names.
  - Quantitative: grades.
- Relationships:
  - Each student → distribution of grades.

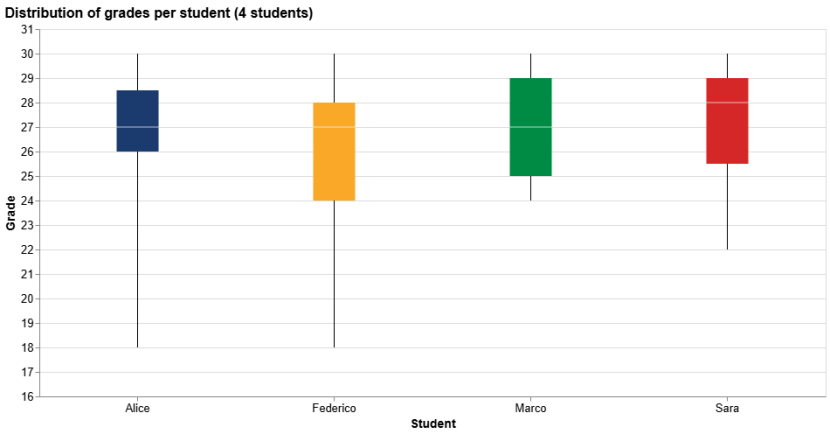


Figure 3: Box Plot for Group.

- Visual Variables / Variations:
  - Position (x-axis): nominal, student.
  - Position (y-axis): quantitative, grade.
  - Color: nominal, distinct per student.
  - Size / Shape: boxplot elements encode distribution statistics.

1.2.2 Temporal Trend of Group Grades

- Visual Task: Trend / Change over Time for multiple entities.
- Type of Data:
  - Temporal: exam dates.
  - Quantitative: grades, CFU (point size).
  - Nominal: student names, honors (30L).
- Relationships:
  - Each student → sequence of grades over time.
  - CFU → point size.
  - Honors → point shape (triangle for 30L, circle otherwise).

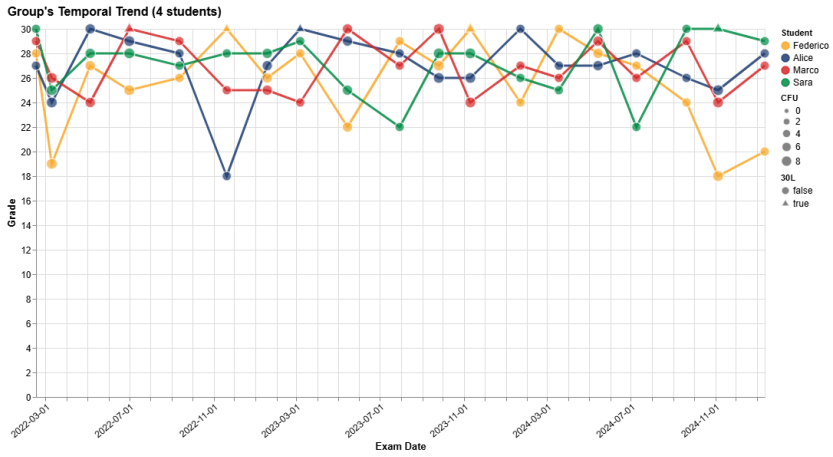


Figure 4: Line Chart for Group Grades.

- Visual Variables / Variations:
  - Position (x-axis): temporal, exam date.
  - Position (y-axis): quantitative, grade.
  - Color: nominal, student-specific.
  - Shape: honors (triangle for 30L, circle otherwise).
  - Size: quantitative (CFU).
  - Line / Point Overlay: shows individual temporal trends per student.

2 Usual Insight

2.1 Wealth and Happiness

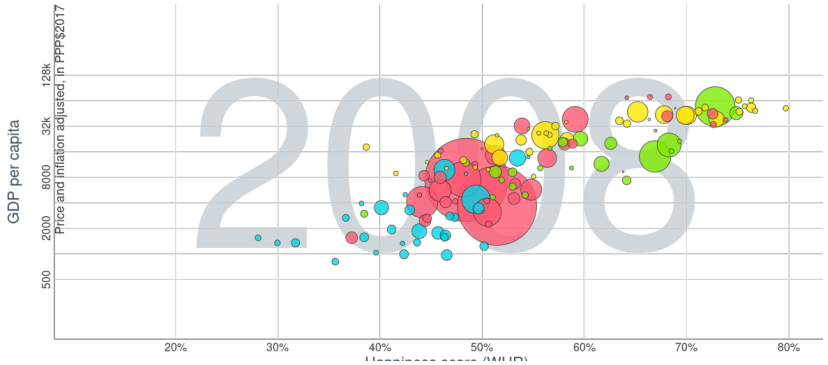


Figure 5: Wealth and Happiness Chart.

As expected, this chart shows a positive and usual correlation: countries with higher GDP per capita (higher on the Y-axis) also report higher Happiness Scores (X-axis).

$$\text{GDP per capita (Wealth)} \uparrow \implies \text{Happiness Score} \uparrow$$

This result is expected (usual) because it is intuitive to think that national wealth provides citizens with a better quality of life, through superior access to healthcare, education, safety, and stability. These factors form the basis for greater well-being and, therefore, a higher happiness score.

2.2 Happiness and Work-Life Balance

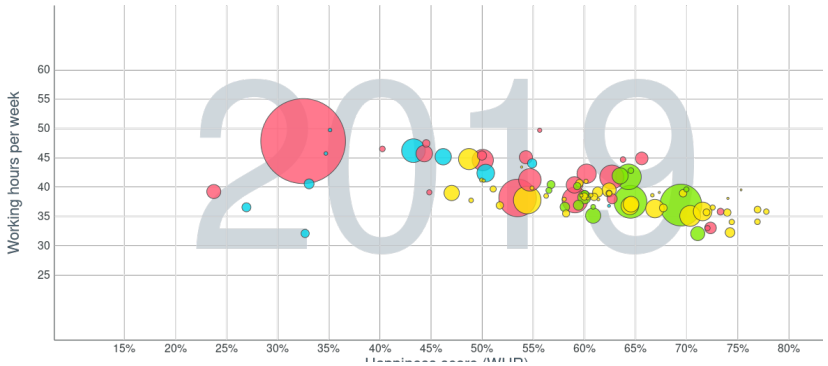


Figure 6: Happiness and Work-Life Balance Chart.

This chart confirms a usual intuition: the happiest countries are those where people work less. A clear negative trend is observed: as the Happiness Score (X-axis) increases, Weekly Working Hours (Y-axis) decrease.

$$\text{Happiness Score} \uparrow \implies \text{Weekly Working Hours} \downarrow$$

This result is expected (usual) because work-life balance is a crucial factor for well-being. Having more free time for family, hobbies, and rest, thanks to shorter working hours, is directly linked to greater personal satisfaction and, consequently, a higher happiness score.

2.3 Poverty as an Obstacle to Education

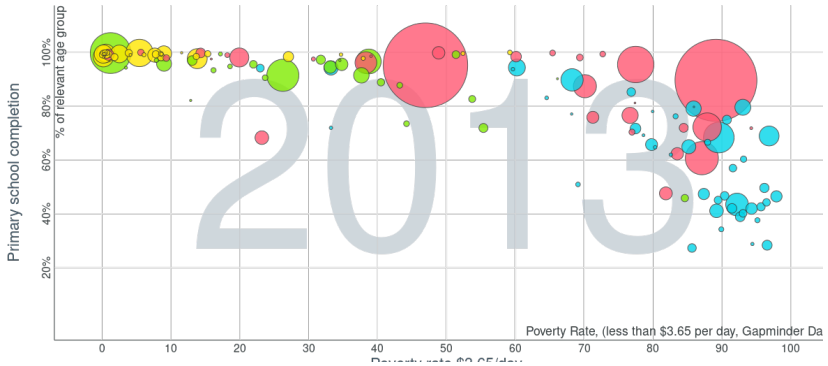


Figure 7: Poverty as an Obstacle to Education Chart.

This chart shows a strong and usual negative correlation: as the Poverty Rate (X-axis) increases, the Primary School Completion Rate (Y-axis) drops sharply.

$$\text{Poverty Rate} \uparrow \implies \text{Primary School Completion} \downarrow$$

This is expected because countries with low poverty (on the left) have almost universally achieved full school completion (close to 100%). In countries with high poverty (on the right), the need to work, lack of infrastructure, and limited family resources prevent many children from completing primary education.

3 Unusual Insight

3.1 The Fertility Paradox

The chart shows a striking and deeply counterintuitive correlation.

$$\text{Wealth (GDP per capita)} \uparrow \implies \text{Number of children (Fertility)} \downarrow$$

The analysis highlights a paradox: one might expect that higher economic availability and security (being richer) would encourage or enable larger families. The data, however, show the exact opposite.

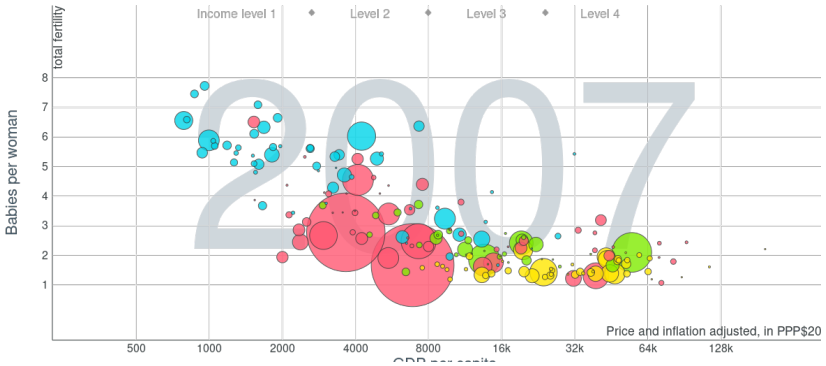


Figure 8: The Fertility Paradox Chart.

Countries with the highest incomes (Level 3 and Level 4) almost universally have lower fertility rates, often below the replacement level (≈ 2.1), while the highest rates are found in countries with lower incomes (Level 1).

3.2 The Work and Wealth Paradox

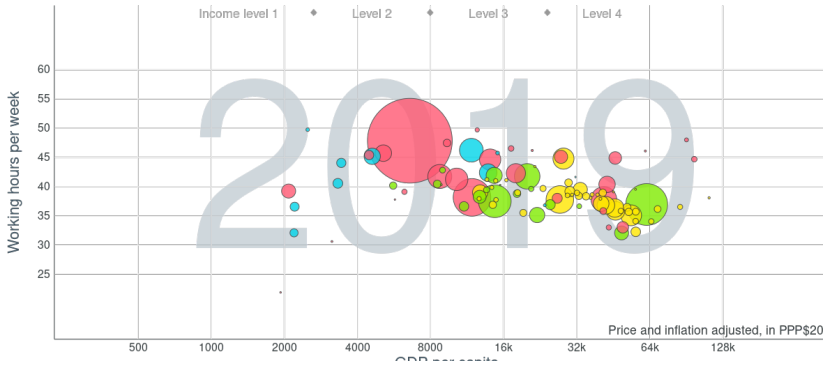


Figure 9: The Work and Wealth Paradox Chart.

This chart shows a striking and counterintuitive correlation: wealthier countries (with higher GDP per capita, towards the right) are those where citizens work, on average, fewer hours per week (lower on the Y-axis).

$$\text{GDP per capita} \uparrow \implies \text{Weekly Working Hours} \downarrow$$

This pattern is unexpected because it challenges common assumptions. Usually, national wealth is associated with working a lot. This chart, however, suggests that wealth does not come from the quantity of hours worked, but from high productivity (the value created per hour), which allows rich countries to generate more GDP in less time.

3.3 The Poverty and the Number of Suicide

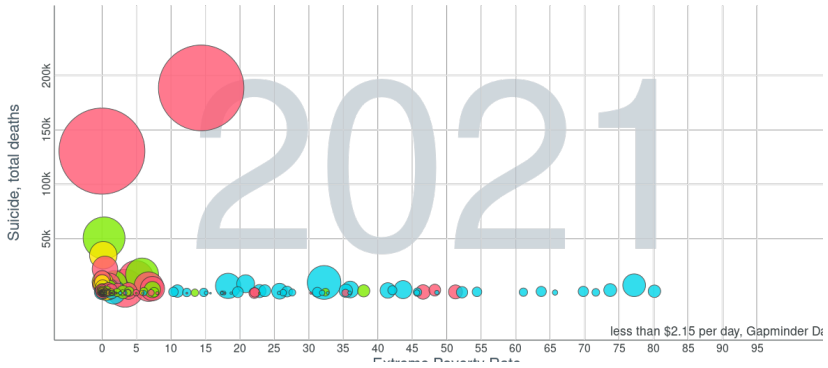


Figure 10: The Poverty and the Number of Suicide Chart

Contrary to what one might expect, this chart shows that there is no direct correlation between being an extremely poor country (X-axis) and having a high total number of suicides (Y-axis).

This is striking because common intuition links economic despair to the tragic decision of suicide. However, the data show that many wealthy countries (with poverty nearly at 0) have total suicide numbers higher than almost all the poorest countries in the world. The determining factor in the chart appears to be population size (as with the large bubbles representing China and India), not the extreme poverty rate.